

☒ 1. Overview of Forensics Tools

Digital forensics tools are used to:

- Collect and **preserve** evidence
- Perform **analysis** on disk, memory, and network data
- Recover deleted files
- Extract and visualize **artifacts** like browser history, registry keys, etc.
- Maintain **integrity** of evidence with hashing and chain of custody features

Forensic tools are broadly categorized as:

Category	Examples
Disk Forensics	FTK, EnCase, Autopsy, OSF
Memory Forensics	Volatility, Rekall
Network Forensics	Wireshark, NetworkMiner
Registry Analysis	RegRipper, Registry Explorer
General System Utilities	Sysinternals Suite, NirSoft tools
Hex Editors	HxD, WinHex, 010 Editor

❏ 2. Sysinternals Suite

❏ Developed by: Microsoft (originally by Mark Russinovich)

The **Sysinternals Suite** is a collection of powerful **Windows utilities** used for diagnostics, incident response, and live forensics.

❏ Key Tools in Sysinternals:

Tool	Function
Process Explorer	Detailed view of running processes (better than Task Manager)
Autoruns	Shows startup programs, drivers, scheduled tasks
PsExec	Execute commands remotely
TCPView	Lists all open network connections
Handle.exe	View which processes have which files open
RAMMap	Shows detailed memory usage

☒ Forensics Use Cases:

- Investigating suspicious processes or malware
- Dumping memory of a running process

- Mapping network ports to processes
- Monitoring unauthorized startup programs
- Live system inspection without installing heavy tools

☒ 3. FTK (Forensic Toolkit)

☒ Developed by: AccessData

FTK is a **comprehensive GUI-based forensic analysis tool** used by law enforcement, corporate investigators, and auditors.

☒ Key Features:

- Full disk and partition analysis

- File carving (recover deleted files)
- Registry analysis
- Email parsing and indexing
- Timeline and hash analysis
- Custom keyword searching
- Strong reporting features

☒ Forensics Use Cases:

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Disk analysis (NTFS, FAT32, exFAT)

- Case management for investigations
 - Recover **email attachments**, artifacts, and chat histories
 - Perform **file signature mismatch detection**
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☒ 4. FTK Imager

FTK Imager is a **lightweight imaging and preview tool** provided by AccessData, commonly used for creating **forensic images** of disks and memory.

☒ Key Features:

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Create **bit-by-bit forensic images** in multiple formats (E01, AFF, RAW)

- Calculate and verify **hash values** (MD5, SHA1)
- Preview contents **without altering original media**
- Extract specific files or folders
- Capture **RAM dumps** (live memory capture)

☒ Forensics Use Cases:

- Creating forensically sound disk images
- Validating evidence integrity

- Viewing deleted or hidden files
 - Creating portable snapshots of USB drives, CDs, or HDDs
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☒ 5. OSForensics (OSF)

☒ Developed by: PassMark Software

OSF is a feature-rich **forensic suite** that provides multiple tools for investigators in one platform.

☒ Key Features:

- Search files by keyword, hash, or content
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Analyze web browser history and email archives

- File system and registry viewer
- Password cracking
- Timeline visualization
- Disk imaging and drive cloning
- Hash set management

☒ Forensics Use Cases:

- **Triage tool** for rapid field analysis

- Comprehensive **artifact analysis** for Windows systems
 - Hash match with known "bad files" (NSRL)
 - View and recover deleted emails or system files
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☒ 6. Hex Editors (HxD, WinHex, 010 Editor)

☒ What is a Hex Editor?

A **hex editor** allows you to view and edit binary files at the **byte level**. You see contents as hexadecimal values and ASCII characters side-by-side.

☒ Common Tools:

Tool	Notes
HxD	Free, lightweight, easy for quick binary inspection
WinHex	Professional hex editor with forensic functions
010 Editor	Advanced hex templates for structured binary formats

☒ Forensics Use Cases:

- Examine **file headers and footers**
- Detect **file signature mismatches**
- Recover **partially deleted files**
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Identify malicious binary content

- Read slack space and unallocated space manually

☒ Summary Table:

Tool	Best For	Strength
Sysinternals	Live system analysis	Portable and powerful
FTK	Disk, email, registry analysis	Enterprise-grade
FTK Imager	Imaging & preview	Fast and free
OSF	All-in-one forensics	UI-rich, feature-packed
Hex Editors	Byte-level file analysis	Precision & control